Module 6 -- Robotics Projects

Lesson 48: Remote-Controlled Robot (Bluetooth)

★ Star-5 Expert ⏰ 50--60 minutes 👤 Age 6--12 years ⚡ 350 XP

Your Mission Today

Robot Commander, up until now your robot has been running on its own — following lines and dodging obstacles. Today, YOU take the controls! You are going to add a **Bluetooth module** to your robot so you can drive it from your phone. Send commands wirelessly — forward, backward, left, right, stop — and watch your robot obey from across the room. This is wireless remote control, and it is incredibly cool!

Learning Objectives

By the end of this lesson, you will be able to:

- Explain how Bluetooth communication works (in simple terms)
- Wire the HC-05 Bluetooth module to Arduino
- Pair a smartphone with the HC-05
- Write code to receive Bluetooth commands and control the robot
- Switch between manual (phone) and autonomous (obstacle-avoidance) modes
- Use your Magic Measurement Wand to measure HC-05 TX/RX pin voltages

What You Need

Item	Qty
Your robot from Lessons 43-47	1
HC-05 Bluetooth module	1
Jumper wires (female-to-male)	4
1k-ohm resistor	1
2k-ohm resistor (or two 1k-ohm in series)	1
Smartphone with Bluetooth	1
Bluetooth controller app (free)	1
Multimeter (your Magic Measurement Wand!)	1
USB cable (for programming)	1

How to Teach This Lesson

Step 1: Hook — Wireless Magic (5 min)

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"Your robot has been amazing — following lines, dodging obstacles, flashing lights. But what if you could control it from your PHONE? Like a video game, but the character is REAL and driving around your room?"

"Bluetooth is the same technology that connects your wireless headphones to your phone. Instead of sending music, we are going to send COMMANDS. When you press a button on your phone, it sends a letter — F for forward, B for backward, L for left, R for right, S for stop. Your Arduino reads the letter and makes the robot move. It is like sending text messages to your robot!"

```
How Bluetooth Control Works:

[Your Phone]                    [HC-05 Module]         [Arduino]
    |                               |                      |
    |--- "F" (forward) --------->   |--- serial data ---> |
    |                               |                      |
    |                               |                      +--> forward(200)
    |                               |                      |
    |--- "S" (stop) ----------->    |--- serial data ---> |
    |                               |                      |
    |                               |                      +--> stopRobot()
```

Award: **+10 XP** **for understanding Bluetooth communication!**

Step 2: Wiring the HC-05 Module (10 min)

The HC-05 module communicates using serial (TX/RX) just like the USB connection, but wirelessly!

IMPORTANT: The HC-05 runs at 3.3V logic but has a 5V tolerant VCC pin. The TX pin from Arduino is 5V, but the HC-05 RX pin needs 3.3V. We use a **voltage divider** to protect it!

Wiring Diagram:

```
HC-05 Pin    Connection
---------    ----------
VCC     -----> Arduino 5V
GND     -----> Arduino GND
TXD     -----> Arduino Pin 7 (SoftwareSerial RX)
RXD     <----- Voltage Divider from Arduino Pin 8

Voltage Divider for RXD:

Arduino Pin 8 ---[1k ohm]---+---[2k ohm]--- GND
                            |
                        HC-05 RXD

This converts 5V down to ~3.3V to protect the HC-05!
```

Why SoftwareSerial?

> "Arduino only has one hardware serial port (pins 0 and 1), and we need that for the USB connection to upload code. So we use a library called SoftwareSerial to create a SECOND serial port on any pins we choose. It is like adding an extra phone line to your house!"

> "IMPORTANT: Disconnect the HC-05 RXD and TXD wires when uploading code via USB! They can interfere with programming."

Award: **+20 XP** **for wiring the HC-05 with voltage divider!**

Step 3: The Bluetooth Controller App (5 min)

On an Android phone:

1. Search the app store for "Arduino Bluetooth Controller" or "Bluetooth RC Controller"
2. Install a free app that lets you send single character commands

On an iPhone:

- Note: HC-05 works with Android. For iPhone, you need an HM-10 (BLE) module instead.
- Apps: "LightBlue" or "BLE Terminal" work with HM-10

Pairing:

1. Turn on the robot (power the HC-05)
2. The HC-05 LED will blink rapidly — it is waiting to pair
3. On your phone, go to Bluetooth settings
4. Find "HC-05" in the list (default password: 1234 or 0000)
5. Pair with it
6. Open the controller app and connect to HC-05
7. The HC-05 LED will blink slowly — it is now connected!

> 💬
> "When the LED on the HC-05 blinks FAST, it is lonely and looking for a friend. When it blinks SLOW, it found your phone and they are talking!"

Award: **+10 XP** **for successfully pairing your phone with the HC-05!**

Step 4: Bluetooth Control Code (12 min)

```
// Lesson 48: Bluetooth Remote-Controlled Robot
#include <SoftwareSerial.h>

// Bluetooth serial on pins 7 (RX) and 8 (TX)
SoftwareSerial bluetooth(7, 8);  // RX, TX

// Motor pins (same as before)
int leftENA = 9;
int leftIN1 = 2;
int leftIN2 = 3;
int rightENB = 10;
int rightIN3 = 4;
int rightIN4 = 5;

// LED and buzzer pins
int greenLeft = A1;
int greenRight = A2;
int redLeft = A3;
int redRight = A4;
int blueStatus = 13;
int buzzerPin = A5;

int driveSpeed = 200;

void setup() {
  // Motor pins
  pinMode(leftENA, OUTPUT);
  pinMode(leftIN1, OUTPUT);
  pinMode(leftIN2, OUTPUT);
  pinMode(rightENB, OUTPUT);
  pinMode(rightIN3, OUTPUT);
  pinMode(rightIN4, OUTPUT);

  // LED pins
  pinMode(greenLeft, OUTPUT);
  pinMode(greenRight, OUTPUT);
  pinMode(redLeft, OUTPUT);
  pinMode(redRight, OUTPUT);
  pinMode(blueStatus, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  // Start both serial connections
```

```
  bluetooth.begin(9600);  // HC-05 default baud rate

  Serial.println("Bluetooth Robot Ready!");
  Serial.println("Waiting for connection...");

  // Startup blink
  for (int i = 0; i < 3; i++) {
    digitalWrite(blueStatus, HIGH);
    delay(200);
    digitalWrite(blueStatus, LOW);
    delay(200);
  }
}

void loop() {
  if (bluetooth.available()) {
    char command = bluetooth.read();

    // Show command on Serial Monitor too
    Serial.print("Received: ");
    Serial.println(command);

    // Process command
    switch (command) {
      case 'F':  // Forward
      case 'f':
        forward(driveSpeed);
        digitalWrite(greenLeft, HIGH);
        digitalWrite(greenRight, HIGH);
        digitalWrite(redLeft, LOW);
        digitalWrite(redRight, LOW);
        break;

      case 'B':  // Backward
      case 'b':
        backward(driveSpeed);
        digitalWrite(greenLeft, LOW);
        digitalWrite(greenRight, LOW);
        digitalWrite(redLeft, HIGH);
        digitalWrite(redRight, HIGH);
        break;

      case 'L':  // Turn Left
      case 'l':
        turnLeft(driveSpeed);
```

```
        break;

      case 'R':  // Turn Right
      case 'r':
        turnRight(driveSpeed);
        digitalWrite(greenLeft, LOW);
        digitalWrite(greenRight, HIGH);
        break;

      case 'S':  // Stop
      case 's':
        stopRobot();
        digitalWrite(greenLeft, LOW);
        digitalWrite(greenRight, LOW);
        digitalWrite(redLeft, LOW);
        digitalWrite(redRight, LOW);
        digitalWrite(blueStatus, HIGH);
        break;

      case 'H':  // Horn!
      case 'h':
        tone(buzzerPin, 800, 300);
        delay(350);
        noTone(buzzerPin);
        break;

      case '1':  // Speed: Slow
        driveSpeed = 120;
        Serial.println("Speed: SLOW");
        break;

      case '2':  // Speed: Medium
        driveSpeed = 180;
        Serial.println("Speed: MEDIUM");
        break;

      case '3':  // Speed: Fast
        driveSpeed = 240;
        Serial.println("Speed: FAST");
        break;
    }
  }
}

// === MOVEMENT FUNCTIONS ===
```

```
  digitalWrite(leftIN2, LOW);
  digitalWrite(rightIN3, HIGH);
  digitalWrite(rightIN4, LOW);
  analogWrite(leftENA, speed);
  analogWrite(rightENB, speed);
}

void backward(int speed) {
  digitalWrite(leftIN1, LOW);
  digitalWrite(leftIN2, HIGH);
  digitalWrite(rightIN3, LOW);
  digitalWrite(rightIN4, HIGH);
  analogWrite(leftENA, speed);
  analogWrite(rightENB, speed);
}

void turnLeft(int speed) {
  digitalWrite(leftIN1, LOW);
  digitalWrite(leftIN2, HIGH);
  digitalWrite(rightIN3, HIGH);
  digitalWrite(rightIN4, LOW);
  analogWrite(leftENA, speed);
  analogWrite(rightENB, speed);
}

void turnRight(int speed) {
  digitalWrite(leftIN1, HIGH);
  digitalWrite(leftIN2, LOW);
  digitalWrite(rightIN3, LOW);
  digitalWrite(rightIN4, HIGH);
  analogWrite(leftENA, speed);
  analogWrite(rightENB, speed);
}

void stopRobot() {
  digitalWrite(leftIN1, LOW);
  digitalWrite(leftIN2, LOW);
  digitalWrite(rightIN3, LOW);
  digitalWrite(rightIN4, LOW);
  analogWrite(leftENA, 0);
  analogWrite(rightENB, 0);
}
```

Command Reference Card:

```
+------+------------------+
| Key  | Action           |
+------+------------------+
| F    | Forward          |
| B    | Backward         |
| L    | Turn Left        |
| R    | Turn Right       |
| S    | Stop             |
| H    | Horn (beep!)     |
| 1    | Slow speed       |
| 2    | Medium speed     |
| 3    | Fast speed       |
+------+------------------+
```

Award: **+40 XP** **for uploading and testing Bluetooth control!**

Step 5: Test Drive! (5 min)

"Time to take your robot for a spin! Open the Bluetooth app on your phone, connect to HC-05, and start driving!"

Test Checklist:

```
| Test                      | Works? |
|---------------------------|--------|
| Forward command (F)       |        |
| Backward command (B)      |        |
| Turn Left (L)             |        |
| Turn Right (R)            |        |
| Stop (S)                  |        |
| Horn (H)                  |        |
| Speed change (1, 2, 3)    |        |
| Green LEDs on forward     |        |
| Red LEDs on backward      |        |
```

"Drive it around the room! Try navigating around furniture. Can you drive it into the kitchen and back?"

Award: **+30 XP** **for a successful Bluetooth test drive!**

Step 6: Bonus — Dual Mode (Bluetooth + Autonomous) (5 min)

What if your robot could switch between phone control and autonomous mode?

```
// Add this to your code:
bool autonomousMode = false;

// In the loop(), add these commands:
case 'A':  // Switch to Autonomous mode
case 'a':
  autonomousMode = true;
  Serial.println("AUTONOMOUS MODE!");
  tone(buzzerPin, 1000, 200);
  break;

case 'M':  // Switch to Manual mode
case 'm':
  autonomousMode = false;
  stopRobot();
  Serial.println("MANUAL MODE!");
  tone(buzzerPin, 500, 200);
  break;

// At the top of loop():
if (autonomousMode) {
  // Run obstacle-avoidance code here
  // (from Lesson 46)
  return;  // Skip Bluetooth command reading
}
```

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"Now press A on your phone and the robot drives itself! Press M to take back control. You just built a robot with TWO brain modes!"

Award: **+20 XP** **for implementing dual mode!**

Step 7: Wand Check — HC-05 Voltages (8 min)

"Let us use your Magic Measurement Wand to peek at the Bluetooth communication signals!"

Set the Wand to **DC Volts**.

Measurement 1: HC-05 Power

```
| Measurement        | Expected | Your Reading |
|--------------------|---------|-------------|
| HC-05 VCC pin      | 5.0V     |             |
| HC-05 GND pin      | 0V       |             |
```

Measurement 2: TX/RX Pin Voltages

```
| Measurement                  | Expected     | Your Reading |
|------------------------------|-------------|-------------|
| HC-05 TXD pin (idle)        | ~3.3V        |             |
| HC-05 RXD pin (idle)        | ~3.3V        |             |
| Voltage divider output      | ~3.0--3.4V    |              |
| Arduino Pin 8 (before divider) | ~5.0V    |             |
```

"Look at that! The Arduino sends 5V, but our voltage divider drops it to 3.3V before it reaches the HC-05. Without that voltage divider, we could damage the HC-05. Your Wand just confirmed the protection is working!"

Measurement 3: During Communication

Send a few commands from your phone and watch the Wand readings on the TXD pin:

"When data is being sent, the voltage on the TX pin flickers rapidly between 0V and 3.3V. Each flicker is a binary digit — a 1 or a 0 — that makes up the letter you sent. Your Wand is seeing digital data in real time!"

Award: **+50 XP** **for all Bluetooth Wand measurements!**

Quick Quiz — Earn Bonus XP!

Question 1: Why do we need a voltage divider between Arduino TX and HC-05 RX?

- A) To make the signal louder
- B) To drop 5V to 3.3V so we do not damage the HC-05
- C) To make the robot go faster

(Correct: B — +20 XP !)

Question 2: What does the HC-05 LED blinking pattern tell you?

- A) Fast blink = connected, slow blink = searching
- B) Fast blink = searching for connection, slow blink = connected
- C) Blinking means the battery is low

(Correct: B — +20 XP !)

Question 3: What library lets us create extra serial ports on Arduino?

- A) Servo.h
- B) SoftwareSerial.h
- C) Bluetooth.h

(Correct: B — +20 XP !)

Question 4: What is the default pairing password for most HC-05 modules?

- A) password
- B) 1234
- C) bluetooth

(Correct: B — +20 XP !)

Lesson 48 Complete!

```
==========================================

   WIRELESS COMMANDER BADGE UNLOCKED!

   Skills unlocked:
   [check] Wired HC-05 with voltage divider
   [check] Paired phone with Bluetooth
   [check] Controlled robot wirelessly
   [check] Added speed control and horn
   [check] Implemented dual mode (manual/auto)
   [check] Measured Bluetooth voltages with Wand

==========================================
```

XP Breakdown:

Activity	XP
Hook (Bluetooth understanding)	10
HC-05 wiring with voltage divider	20
Phone pairing	10
Bluetooth control code	40
Successful test drive	30
Dual mode bonus	20
Wand Check (all measurements)	50
Quiz (4 questions)	80
TOTAL POSSIBLE	**260**

Coming Up Next...

In **Lesson 49**, you are going to build a **robot arm**! Using servo motors and popsicle sticks, you will create a 3-joint arm that can pick up and move small objects. You will control each joint with a potentiometer — like a joystick for a crane!

Troubleshooting

Problem	Fix
Cannot find HC-05 in phone Bluetooth list	Make sure HC-05 is powered on (LED blinking fast)
Wrong password	Try both 1234 and 0000
Connected but no response to commands	Check RX/TX wires are not swapped
Robot moves erratically	Check baud rate matches (9600 in both code and module)
Cannot upload code with HC-05 connected	Disconnect HC-05 TX/RX wires during upload
HC-05 LED does not blink	Check VCC and GND connections
Voltage divider reads wrong voltage	Check resistor values — 1k and 2k (or two 1k in series)